

Assignment #4 (RSA Public-Key Generation)

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Solution

Here is a function in Python to public/private key pair for RSA.

```
def find_d(p, q, e):
    n = p * q
    phi = (p - 1) * (q - 1)

    d = 0
    while True:
        d = d + 1

    if d > n:
        print("error")
        return None

    x = (d * e) % phi

    if x == 1:
        print("d =", d)
        return d

find_d(19, 31, 37) # 73
find_d(19, 31, 17) # Output 413
find_d(19, 31, 5) # error
```

Can not generate a key for $e = 5$, because a valid RSA encryption key must be **coprime with $\phi(n)$** , meaning **$\gcd(e, \phi(n)) = 1$** . With typical values like **$p = 11$ and $q = 13$** , we get **$\phi(n) = 120$** , and since **$\gcd(5, 120) = 5$** , the modular inverse of 5 modulo 120 does **not** exist. As a result, the equation **$e \cdot d \equiv 1 \pmod{\phi(n)}$** has no solution, so the loop never finds a valid **d** and correctly terminates with an error.